

Dr. Divyesh A. Mistry

Academic Curriculum Vitae | Computational Materials Science and Multiscale Mechanics

d.mistryg@gmail.com | Website | LinkedIn | GitHub | Google Scholar

RESEARCH PROFILE

Research focuses on mechanistic multiscale descriptions of plasticity, dislocation evolution and fracture in advanced structural materials. Current and recent work combines atomistic simulation, discrete dislocation dynamics, continuum mechanics and microstructure generation to study nickel-based superalloys and refractory tungsten alloys.

APPOINTMENTS

Visiting Researcher, Computational Materials Science

Merrimack College, North Andover, MA, USA | Apr 2026 - Jul 31, 2026

- Investigated temperature-dependent crack-tip instability and dislocation-mediated relaxation in tungsten and tungsten-rhenium alloys using MD and DDD.
- Developed reproducible LAMMPS, Python and polycrystalline DDD workflows for high-performance computing environments.
- Contributed to technical reports, research proposals and student mentoring.

Junior Research Fellow, DRDO Project

Indian Institute of Technology Bombay | Jan 2025 - Dec 2025

- Investigated constitutive behavior and high-strain-rate response of shape-memory-alloy-based composites using impact experiments and numerical simulation.

Ph.D. Research Scholar

Indian Institute of Technology Bombay | Jan 2019 - Jun 2025

- Developed atomistic and mesoscale models of dislocation-mediated deformation in nickel-based superalloys.

Graduate Teaching Assistant

Indian Institute of Technology Bombay | Jan 2019 - Dec 2023

Assistant Professor, Mechanical Engineering

CMR Institute of Technology, Bengaluru, India | Jun 2016 - Dec 2018

Graduate Teaching Assistant and M.Tech. Researcher

Indian Institute of Technology Kharagpur | Jun 2014 - Jul 2016

EDUCATION

Ph.D., Aerospace Engineering

Indian Institute of Technology Bombay | 2019 - 2025

- Thesis: Multiscale Modeling of Prior Particle Boundaries in Nickel-based Superalloys.

M.Tech., Mechanical Engineering - Mechanical Systems Design

Indian Institute of Technology Kharagpur | 2014 - 2016

B.E., Aeronautical Engineering

The Aeronautical Society of India | 2010 - 2013

JOURNAL ARTICLES

A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-based Superalloys

Divyesh A. Mistry, Tawqeer Nasir Tak, R. Sankarasubramanian and P. J. Guruprasad

Journal of Engineering Materials and Technology, published online 2026. DOI: 10.1115/1.4072448

Size-dependent Power Laws for Edge Dislocations in Nickel Superalloys: A Molecular Dynamics Study

Divyesh A. Mistry and Amuthan A. Ramabathiran

Computational Materials Science 259 (2025), 114122. DOI: 10.1016/j.commatsci.2025.114122

MANUSCRIPTS AND OPEN RESEARCH SOFTWARE

Multiscale fracture and collective dislocation evolution in tungsten-based materials

Divyesh A. Mistry and collaborators

Journal manuscript in preparation.

Polycrystalline Tungsten Discrete Dislocation Dynamics Workflow

Dr. Divyesh A. Mistry

Open-source modular Python workflow with examples, tests and citation metadata.

SELECTED CONFERENCE CONTRIBUTIONS

- Multiscale Modeling of Dislocation-Precipitate-PPB Interactions in Powder-Metallurgy Nickel Superalloys, Engineering Mechanics Institute Conference, Boulder, Colorado, USA, 2026.
- Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys, IMPLAST 2025, IIT Roorkee, India.
- An Atomistically Informed Discrete Dislocation Dynamics Study of Prior Particle Boundaries in Ni-Based Superalloys, COMPLAS 2023, Barcelona, Spain.

RESEARCH METHODS AND SOFTWARE

Simulation: Molecular dynamics; discrete dislocation dynamics; microstructure-sensitive and multiscale mechanics; fracture, plasticity and defect evolution.

Tools: LAMMPS, OVITO/DXA, Neper, OpenDiS/ExaDiS workflows, ParaView, Git, Linux/Unix and HPC scheduling environments.

Programming and analysis: Python, NumPy, Pandas, scientific visualization, workflow automation, C/C++ and Fortran.

TEACHING AND MENTORING

- Teaching and laboratory support in computational mechanics, continuum mechanics, material behavior and engineering simulation.
- Mentoring of graduate and undergraduate students in molecular dynamics setup, parallel computation and scientific post-processing.
- Development of reproducible examples and documentation for open-source DDD research software.

AWARDS AND PROFESSIONAL MEMBERSHIPS

- Member, The Minerals, Metals & Materials Society, 2025.
- Institution of Eminence travel support, IIT Bombay, for TMS 2025, USA.
- IIT Bombay travel grant for COMPLAS 2023, Spain.
- MHRD Teaching Assistantship through research project, IIT Bombay, 2019.
- Graduate Aptitude Test in Engineering (GATE) 2014: All India Rank 69.
- Permanent associate member, Aeronautical Society of India, 2013.

SELECTED CERTIFICATIONS

- Using Python for Research, HarvardX, 2020.
- The Unix Workbench, Johns Hopkins University, 2020.
- Material Behavior, Georgia Institute of Technology, 2020.
- A Hands-on Introduction to Engineering Simulations, CornellX, 2018.